Ethernet kommutator asoslari

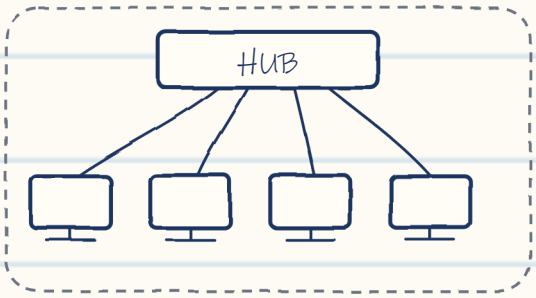
Ethernet Switch Fundamentals

Asosiy atamalar
- Hub — hamma portga takrorlaydi, bitta kolliziya domeni
- Kolliziya domeni — bir vaqtda bitta kadr yuradigan hudud
- MAC jadval — «MAC manzil → port» moslik ro'yxati
- Flood — kadrni qolgan hamma portga yuborish
- Broadcast domen — broadcast kadr yetib boradigan hudud

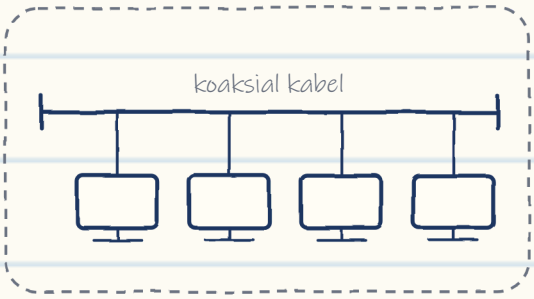
1. Umumiy (shared) Ethernet

Hub yoki koaksial kabel — hammasi bitta kolliziya domeni.

- to'qnashuvni CSMA/CD aniqlaydi, kadr qayta yuboriladi
- host ko'paysa — kolliziya ko'p, unum tushadi
- polosa bo'linadi, har kadrni hamma oladi



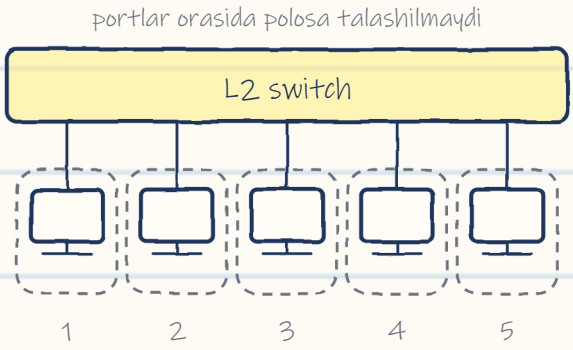
bitta kolliziya domeni



bitta kolliziya domeni

2. Kommutatsiyalangan Ethernet

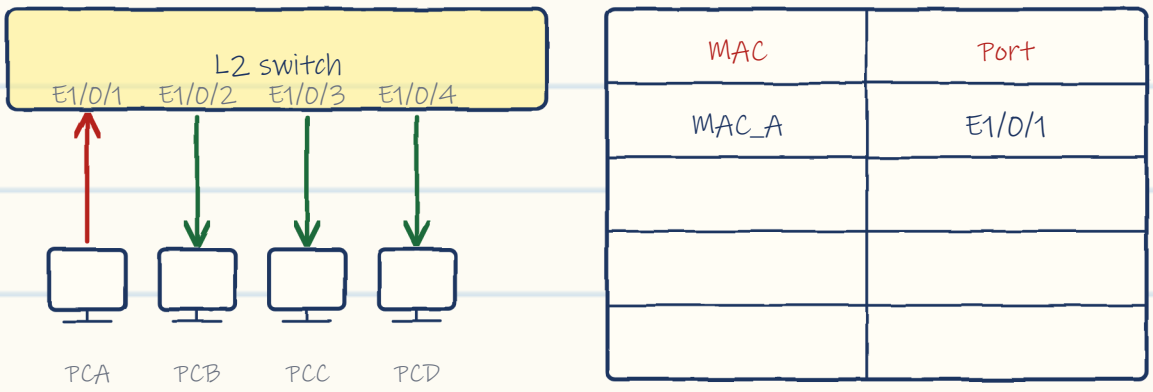
L2 kommutator — IEEE 802.1D ko'prik (bridge) vorisi: ko'p port, tez uzatish.



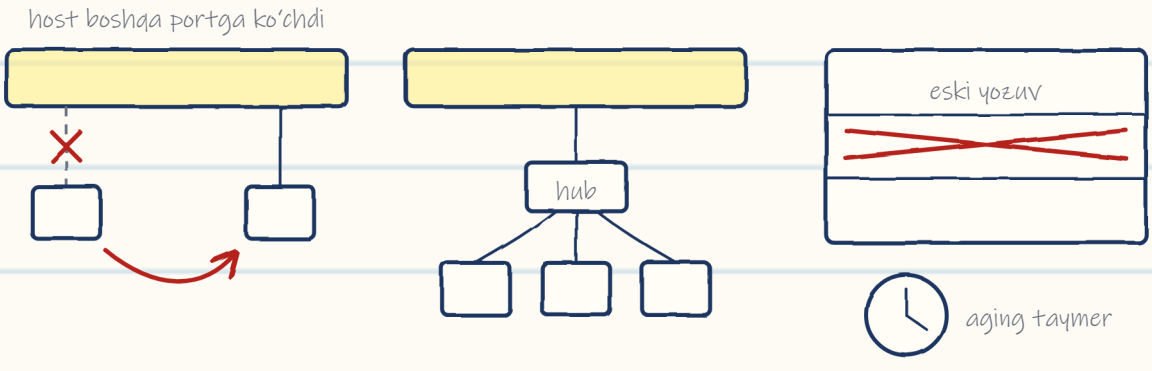
har port — alohida kolliziya domeni

3. MAC jadvalini o‘rganish

Boshda jadval bo‘sh. Kadr kelganda manba MAC + kirish porti yoziladi, manzil topilmasa — hamma portga tarqatiladi.



1-rasm. PCA yubordi → MAC_A yozildi, kadr qolgan portlarga tarqaldi



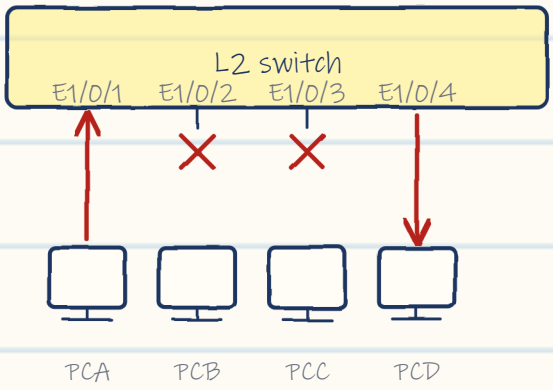
bitta MAC — bitta port

bir portda ko‘p MAC

yangilanmasa — o‘chadi

4. Unicast kadrni uzatish

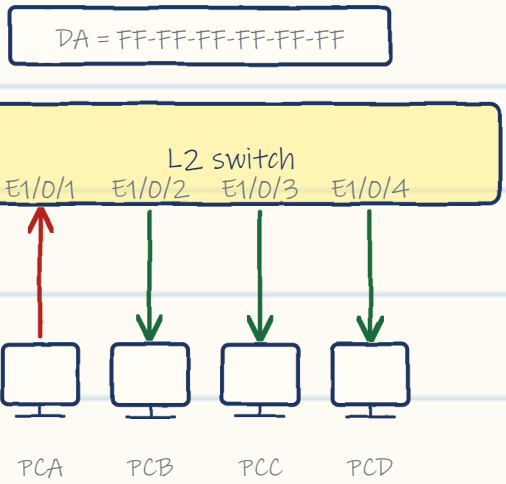
Manzil MAC jadvalda bor — kadr faqat o'sha portga chiqadi.



MAC	Port
MAC_A	E1/0/1
MAC_B	E1/0/2
MAC_C	E1/0/3
MAC_D	E1/0/4

2-rasm. PCA → PCD: jadvalda MAC_D bor, kadr faqat E1/0/4 ga

5. Flood — hamma portga



flood bo'ladi:

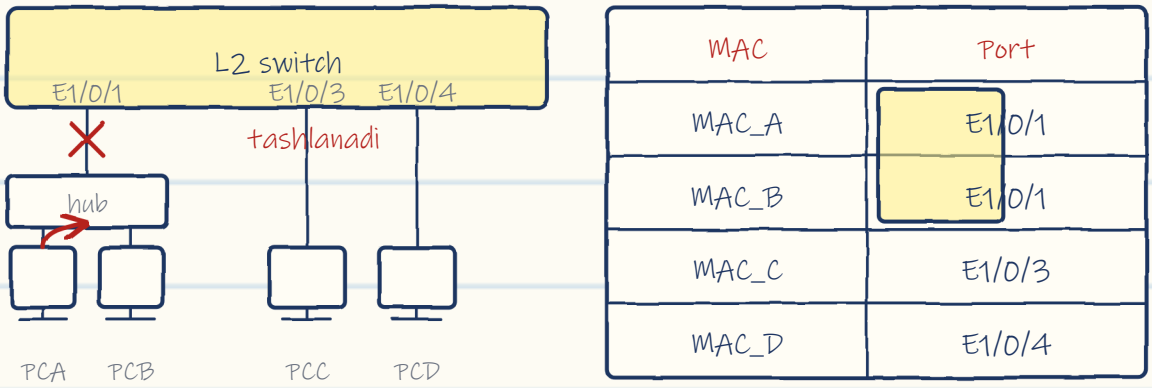
broadcast · multicast

noma'lum unicast

kelgan portdan tashqari hammasiga

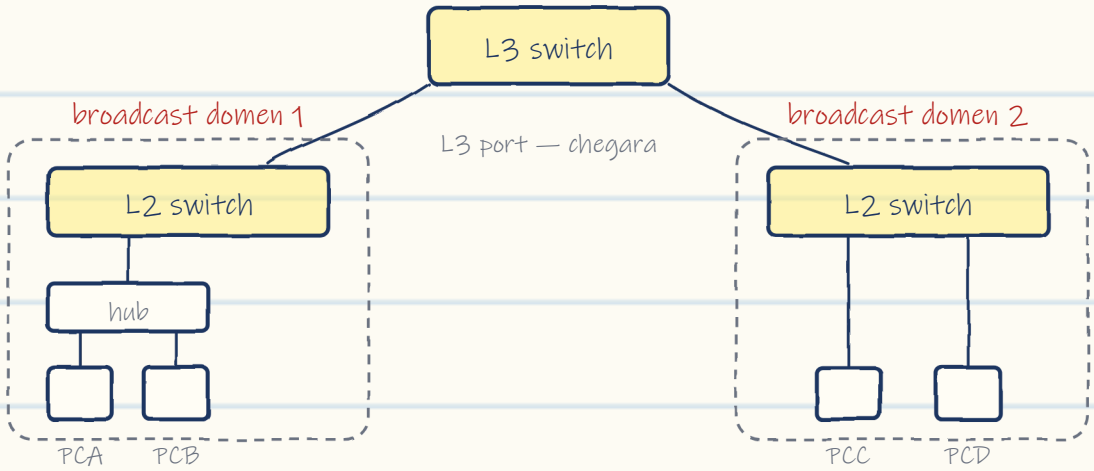
6. Filtrlash

Manzil kelgan portning o'zida o'rganilgan bo'lsa — kadr tashlanadi.

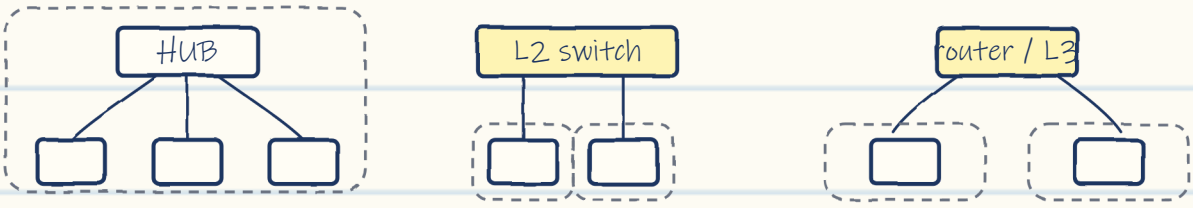


3-rasm. PCA → PCB: ikkalasi ham E1/0/1 da — uzatilmaydi

7. Broadcast domen



8. Qurilmalarni taqqoslash



1 kolliziya domeni
1 broadcast domen

har port — o'z domeni
1 broadcast domen

broadcast chegarasi
2 broadcast domen

4-rasm. Kolliziyani kommutator bo'ladi, broadcastni — L3

	Kolliziya domeni	Broadcast domeni
Hub	bitta	bitta
L2 kommutator	har port — alohida	bitta
Router / L3 port	har port — alohida	har port — alohida

Xulosa

- Hub — fizik qatlam, bitta kolliziya domeni, CSMA/CD
- L2 kommutator — kanal qatlami, IEEE 802.1D, har port alohida domen
- Jadval manba MAC bo'yicha to'ladi, aging bilan tozalanadi
- Topildi — bitta portga · topilmadi yoki broadcast — flood · o'sha portda — tashlanadi

Tekshirish

1. Kommutator qaysi qatlamda? — kanal (L2)
2. Jadval nima bo'yicha to'ladi? — manba MAC
3. Broadcast MAC? — FF-FF-FF-FF-FF-FF
4. Broadcast domenni kim bo'ladi? — router / L3 port, VLAN